

[DF-4018] [AIS] Credit Profiling for Retail - Migration Created: 25/Dec/25 Updated: 07/Jun/26

Status:	UAT IN PROGRESS
Project:	Digital Factory
Component/s:	 INTEGRATIONS-Initiative
Affects Version/s:	None
Fix Version/s:	None

Type:	Epic	Priority:	P1 - Highest
Reporter:	Abrar S. Bawazir	Assignee:	John Refaat
Resolution:	Unresolved	Votes:	0
Labels:	None		
Σ Remaining Estimate:	Not Specified	Remaining Estimate:	Not Specified
Σ Time Spent:	Not Specified	Time Spent:	Not Specified
Σ Original Estimate:	Not Specified	Original Estimate:	Not Specified

Attachments:	 Copy of Credit Scoring 360 Risk Assessment Results 2025 OB.xlsx  Credit Scoring For Retail.xlsx  Credit Scoring Testing Results.xlsx  Enhanced Credit Risk Assessment API - Testing.postman_collection.json  TRX_fixed.xlsx														
Issue Links:	Polaris issue link implements DF-4005 Grow total successful OB API calls to... To Do Relates relates to DF-5053 [Dev] Calculation are Wrong UAT IN PROGRESS														
Sub-Tasks:	<table><tr><th>Key</th><th>Summary</th><th>Type</th><th>Status</th><th>Assignee</th></tr><tr><td>DF-5053</td><td>[Dev] Calculation are Wrong</td><td>Sub-task</td><td>UAT IN PROGRESS</td><td>Hamza Afandi</td></tr></table>					Key	Summary	Type	Status	Assignee	DF-5053	[Dev] Calculation are Wrong	Sub-task	UAT IN PROGRESS	Hamza Afandi
Key	Summary	Type	Status	Assignee											
DF-5053	[Dev] Calculation are Wrong	Sub-task	UAT IN PROGRESS	Hamza Afandi											
Epic Name:	SmartRisk Retail: Credit Risk Assessment														
Product:	Open Banking New Engine														
Product Owner:	 Abrar S. Bawazir														
Project Manager:	 John Refaat														
Start date:	17/Feb/26														
End Date:	21/Jun/26														
Quarter:	Q2 2026														
Efforts:	M														
Product Pillar:	Open Banking - AIS														
GRC Ticket Type:	Standard														
Confirm: Risk Questionnaire Attached:	Yes, I have attached the Risk Questionnaire.														

Description

Retail Credit Scoring Engine (B2C)

1. Objective

Build a configurable retail credit scoring engine that calculates a standardized Credit Score, Risk Class, and Decision Recommendation using predefined formulas and weighted risk factors, with the ability to override weights and thresholds per client.

The engine must return:

- JSON output
- PDF Credit Report

The attached scoring logic and formulas represent the default configuration.

2. Business Value

2.1 For Clients (Banks / Fintechs)

- Faster, automated credit decisioning
- Reduced manual underwriting effort
- Consistent and explainable scoring logic
- Ability to customize risk appetite per client

2.2 For neotek

- Monetizable scoring engine (API-based)
- Scalable across multiple clients with different configurations

3. In Scope

- Per account
- Retail (Individual) credit scoring through OB transaction & parties' endpoints
- Rule-based weighted scoring (for listed factors only)
- Configurable scoring parameters per client (with default values)
- JSON + PDF outputs

5. High-Level Flow

Approved consent

Scoring engine:

1. Applies formulas
2. Applies configured weights
3. Calculates total score

System assigns:

1. Credit Score
2. Risk Class
3. Decision outcome

System generates:

1. JSON response
2. PDF credit report

6. Scoring Logic & Formulas

Overview

Total Score: 0-1000 points
Factors: 7 weighted components
Rating Bands: 10 tiers (AAA to D)

Factor Breakdown

Factor	Weight	Percentage	Formula
1. Age	150 points	15%	Lookup based on age ranges
2. Income	150 points	15%	Median of last 3 months (spike-excluded)
3. Loan Payment	150 points	15%	Last month loan payment amount
4. Surplus Ratio	150 points	15%	(Income - Expenses) / Income
5. Salary Category	150 points	15%	Risk categorization by income level
6. Cash Flow	150 points	15%	Absolute: Income - Expenses
7. Debt Payment Ratio	100 points	10%	Loan Payment / Income
TOTAL	1000 points	100%	Sum of all factors

Metric	Value	Industry Standard
Debt-to-Income Ratio	28.5%	< 35%
Savings Rate	18.2%	> 15%
Surplus Ratio	27.0%	> 20%
Monthly Cash Flow	SAR 4,200	Positive
Emergency Fund	4.5 Months	> 3 Months
Avg Balance (6M)	SAR 38,500	Stable
Negative Balance Days	0 Days	0 Days
Salary Regularity	100%	> 90%

Acceptance Criteria

- Per account
- The client should have a dynamic setup of the factors wights and the ranges per factor with its assigned point
- The client should configure the Industry Standard for the given matrix
- in other matric section within the report if data were insufficient then the metric should be hidden
- In case of any factor data are not returned then the weight assigned to this factor will be disrupted equally to other remaining factors
- Output Formats: **JSON Output & PDF Report**

<https://www.figma.com/make/8ZX0okSt6yXqry7eRGlymF/Redesign-Credit-Analysis-Report?fullscreen=1&t=4wnYbC7G7gMdsKy-1>

Comments

Comment by [Abrar S. Bawazir](#) [14/May/26]

[John Refaat](#) issues have been submitted to tech, please let us know once fixed.

Testing Summary

Total Tests	Passed	Failed	Warnings
21	11 	5 	5 

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Critical Issues

#	Metric	Account	Computed Result	API/PDF Result	Identified Issue	Required Fix
1	Months of Data 	ACC-148	6 Months	7 Months	March 2026 (Income = 0, Expense = SAR 428) exists within the analysis window but was excluded. Zero-income months are currently skipped.	Include all months between dataStartDate and dataEndDate, including months with zero credits.
2	Avg Balance 6M	ALL	SAR 4,289	SAR 0	API balance value depends on external daily balance feed, not OB transaction records. Current calculation uses net flow instead.	Display "Not Available" until external balance feed integration is completed.
3	L3M Median Income (F2 Input)	ACC-148	SAR 3,100	SAR 2,900	Result impacted by Issue #1. Correct L3M values become [3200, 2900, 0] → Median = SAR 2,900. This affects Surplus Ratio, DBR, Savings Rate, and Emergency Fund.	Include zero-income months before calculating the L3M median.
4	Cash Flow for F6 Score	ACC-148	SAR 1,075	SAR 218	Incorrect denominator and formula implementation. Correct formula: (Total Salary Credits – Total All Debits) ÷ n.	Use active months count correctly and include ALL debits, not only living expenses.
5	Income Volatility (CV)	ACC-148	5.21%	41.21%	Zero-income month excluded from volatility calculation.	Compute CV using ALL months in the analysis window, including zero-income months.
6	Expense Volatility (CV)	ACC-148	3.18%	34.79%	Zero-expense/zero-income month excluded from CV logic.	Compute CV using all months in the analysis period.

7	Average Monthly Expenses	ALL	Includes Loan Payments	Excludes Loan Payments	Backend currently separates loan/mortgage repayments from living expenses before expense calculations. And maybe its minor difference caused by floating-point rounding.	Expense metrics should use (Monthly Debits – Loan Payments) while loan payments remain under totalDebtPayments.
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Warnings & Recommendations

#	Metric	Account	Computed Result	API/PDF Result	Observation	Recommendation
1	Savings Rate	ACC-146 ACC-149	12.42% 35.76%	12.0% 36.0%	Minor difference caused by floating-point rounding.	Round avgSurplus to 2 decimal places before Savings Rate calculation.
3	Emergency Fund	ACC-146 ACC-149	1.70 Months	1.99 Months 10.57 Months	Current implementation uses L3M average expenses instead of full-period average expenses.	Use full-period average expenses for Emergency Fund calculation only.
5	F1 Age Score	ALL	0 / 150	0 / 150	Score matched accidentally due to mismatch between request age and profile age.	Validate request age against national ID profile age.
6	F3 Loan Payment Config Band	ACC-146	150 / 150	150 / 150	Per specification, SAR 1–1,500 should map to 130 points, not 150.	Review and reconcile scoring configuration bands with business specifications.

Comment by [John Refaat](#) [30/Apr/26]

Fixes have been deployed, kindly validate and check

Comment by [Abrar S. Bawazir](#) [30/Apr/26]

[John Refaat](#) let us know once ready to start re-test

[Hamzah O. Afandi](#)

Comment by [Abrar S. Bawazir](#) [20/Apr/26]

Dears [John Refaat](#) [Mohammad AlHaddad](#)

please check reported issue need to be fixed and PDF required changes

CC: [Hamzah O. Afandi](#)

Comment by [Hamzah O. Afandi](#) [14/Apr/26]

Dears [John Refaat](#) [Majid A. Alhassan](#)

Please check the below comments on the PDF version:

1. The circle around the score in the first page need to be aligned with the number
2. Remove Recommendation section and keep the overall risk level (check figma updated)
3. In page 2 the total score text box is not aligned with the other text boxes in the shape
4. In page 4 the Verifiable income ratio text box shape needs to match the figma design
5. In page 6 the stable behavior text box needs to be aligned with the shape of the box
6. In page 8 remove the recommendation box completely and keep final score and risk level

Comment by [John Refaat](#) [02/Apr/26]

Dear [Abrar S. Bawazir](#) ,

Please find attached the UAT Postman Collection to start your activity. Let's arrange a meeting Sunday to conduct the session and get you ready.

All the best,

Comment by [Abrar S. Bawazir](#) [26/Mar/26]

[John Refaat](#) Please share the collection to start testing ...

Comment by [Abrar S. Bawazir](#) [09/Mar/26]

[John Refaat](#) as agreed with GRC, we should proceed with development.

SAMA approval (NoC) will be required only for clients who are planning to utilize this product.

few amendments requested by Compliance to remove recommendation wording has been applied into the PDF report.

CC: [Hamzah O. Afandi](#) [Musab S. Alhazme](#) [Ali S. Albarrak](#) [Anas AlBassam](#)

Comment by [Musab S. Alhazme](#) [05/Mar/26]

Dears ,

we are studying the project with product to team to finalizing either to go with SAMA NOC or not.

thanks

Comment by [Abrar S. Bawazir](#) [02/Mar/26]

Dears [Musab S. Alhazme](#) [Ali S. Albarrak](#)

Please refer to the attached responses to the Credit Profiling Risk Assessment Questionnaire (see attached files)

For context, kindly note the following:

1. We are already certified under the Credit Assessment use case, and this submission covers the planned migration of the existing product into our new engine (with Drahim), along with a set of enhancements, most notably the addition of a PDF report.
2. For delivery purposes, the BRD has been split into two separate items due to differing priorities:

- Retail [DF-4018](#)
- Corporate [DF-14](#)

However, this risk assessment should be treated as covering both segments collectively, as they form a single product.

Please do not hesitate to reach out should you require any clarification or additional information.

CC: [Anas AlBassam](#) [Mohammed A. Almathami](#) [Hamzah O. Afandi](#)

Comment by [Musab S. Alhazme](#) [10/Feb/26]

[Hamzah O. Afandi](#) Greetings, Kindly As discussed, kindly find the attached risk assessment. Please let us know once it is finalized

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